Choudhury 2022 meningioma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Malignant cells form a per-patient archipelago. At the best-mixing integration (θ=10) the kNN graph is a single component (iLISI 1.0→3.0) — the scalars a naïve test reads as "integrated" — yet the embedding is organised BY PATIENT: every patient self-clusters locally (self-neighbour enrichment 1.8–7.2×); 3/5 patients cross the 2.5× line (1/3/4 at 7.2/3.4/2.8×) and the two largest (5,6) hold their own territories. Integration strength does not fix it — the enrichment GROWS with θ (patient 1: 7.2→11.9→12.9×), so Harmony over-corrects. Each patient carries the same embryonic modules → inter-patient CNV/clonal heterogeneity (expected for meningioma), not a shared cell-state continuum → do not force a trajectory.

Gate 2 — module geography → N/A

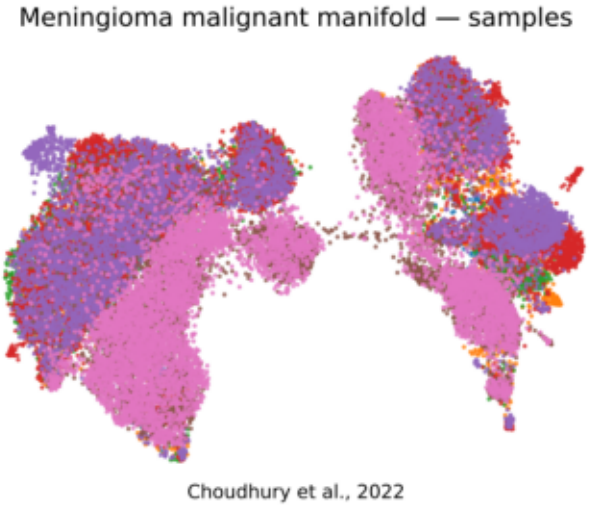
Not reached. Gate 2 tests a Gast/pSom hub and Crest geography on a fitted manifold; with Gate 1 NO-GO no manifold/trajectory is fit, so the module-geography gate does not apply.

Gate 3 — cross-repression → N/A

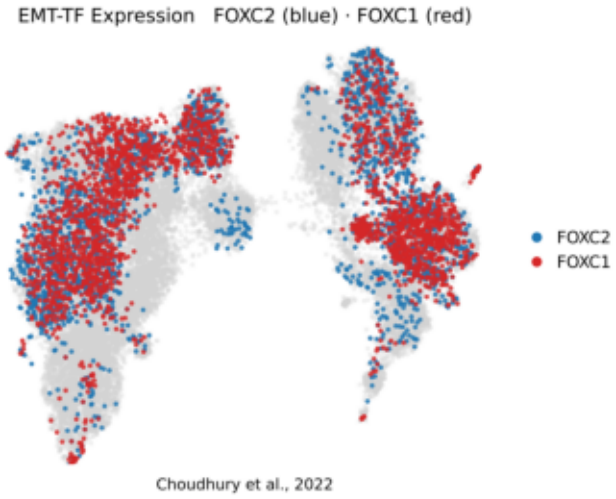
Not reached (branching-only, and downstream of Gate 2). No hub, no bifurcation, no hub r.

STEP FIGURES

Malignant manifold — samples



EMT-TF expression



Dataset.
Meningioma (Choudhury et al., 2022), 10x (UMI counts) — 37,049 malignant cells; 7/10 samples, 5 patients enter preprocessing. First meningioma / first non-glial brain tumour in the atlas.

Verdict.
The embryonic-program pattern does NOT reproduce in Choudhury 2022 meningioma: the malignant compartment is a per-patient archipelago with no sample-integrated, continuous cell-state manifold. Gate 1 NO-GO — analysis stops before the trajectory. A genuine portability NO-GO (integration failure), distinct from a pattern-absent Gate-2 NO-GO. The FOXC mesenchymal TF family is broadly expressed (arachnoid/mesenchymal identity), reported unconditionally without pseudotime.

Parameter choices

Sample threshold	≥780 cells (geo-mean across the 2254→270 gap, 8.35×); keeps 7/10 samples (5 patients)	**Gene / PCA**	2,828 overdispersed (scFates GAM); n_pcs=20 (PC1 alone ≈25% variance, inter-patient/CNV axis)
Integration	Harmony on sample, θ swept 10/20/40; θ=10 best mixing (iLISI 3.0) — diagnostic, not corrective	**Root / geometry**	None — no trajectory fit (Gate 1 NO-GO); geometry not applicable
EMT-TFs shown	FOXC2 (68%) · FOXC1 (67%) — broadest detectable (FOXC/PRRX panel; VIM excluded, not a TF)		